Chaos Theory

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1 Introduction

We can use chaos theory to study errors of the traditional Kalman filter. It is worth, at least for intellectual curiousity, to compute the Lyapunov exponent λ from

$$|\delta\varepsilon(k)| = e^{\lambda k}|\delta\varepsilon(0)|$$

where ε is the error at k. Given the true state and observations follow,

$$x_k = \Phi_k x_{k-1} + A_k u_k + w_k$$

$$y_k = B_k x_k + v_k$$

where x is the true state, Φ is the state-transition model, A is the control input model, u is the control vector, w is the noise, y is the observation of the true state, B is the observation model, v is the observational noise. The Kalman filter predicts the state using

$$\hat{x}_k = \Phi_k \hat{x}_{k-1|k-1} + A_k u_k$$

where $\hat{x}_{k|k}$ is the a posteriori state estimate mean at k from observations up to and including at k. We can subsequently define an error ε as the difference between the predicted state and the true state

$$\varepsilon(k) = x_k - \hat{x}_k$$

$$= \Phi_k x_{k-1} + A_k u_k + w_k - \Phi_k \hat{x}_{k-1|k-1} - A_k u_k = \Phi_k(x_{k-1} - \hat{x}_{k-1|k-1}) + w_k$$

This shows us interestingly that the a posteriori state estimate error from all previous and including observations is 'scaled' by the state-transition model plus noise.

$$\varepsilon(k) = \Phi_k \varepsilon(k-1 \mid k-1) + w_k$$

This gives us a recursive equation for the Lyapunov exponent,

$$|\delta\Phi_k \varepsilon(k-1 \mid k-1) + \delta w_k| = e^{\lambda k}|\delta\Phi_0 \varepsilon(-1 \mid -1) + \delta w_0| \leq |\delta\Phi_k \varepsilon(k-1 \mid k-1)| + |\delta w_0|$$

where, if the states are well-initialized $\varepsilon(-1 \mid -1) = \varepsilon(0) = 0$ giving us

$$|\delta\varepsilon(k)| = e^{\lambda k}|\delta w_0|$$

We are also interested in the derivatives of the error as it tells us about the dynamics of the system over time. Simply

$$\dot{\varepsilon}(k) = \dot{x}_k - \dot{\hat{x}}_k$$

This allows for algebraic simplification dependant on the forms of the derivatives. Below I will show something interesting quickly for the extended Kalman filter (EKF). Something of the following for the time-continuous EKF should yield

$$\dot{\varepsilon} = M\varepsilon$$

which is asymptotically stable if M is a Hurwitz matrix as the system error derivative is negative and ε decreases over time. For the EKF, Φ may be a differentiable function. It is defined by the Jacobian,

$$J_{\Phi_k} = \frac{d\Phi}{dx} \mid_{\hat{x}_{k-1|k-1}, u_k}$$

The system error evolves simply as $\dot{\varepsilon}(k) \approx \Phi_k(\varepsilon)$. Interestingly X describes vectors in the tangent space (of the phase space) and are defined from the Jacobian matrix J_Φ. It satisfies an equation

$$\dot{X} = J_\Phi X$$

Then it can be shown that there exists a matrix Λ such that

$$\Lambda = \lim_{k\to\infty} \frac{\ln(X_k X_k^T)}{2k}$$

where the eigenvalues of this matrix are defined exactly by λ.